

Similarly, the high pressure to the right of the vertical stabilizer generates negative lift on the right semispan of the horizontal stabilizer.

As can be seen from Fig. 5.4.1, the aerodynamic forces produced on the tail of an airplane as a result of sideslip are quite complex, even for this very simple tail geometry. Furthermore, the distribution and magnitude of these forces are affected by the geometry of both the horizontal and vertical stabilizers. When one considers all of the many possible variations in tail geometry, it is not practical to present results that show how the lift slope for the vertical stabilizer varies with all attributes of tail geometry. However, aspect ratio plays a major role in the variation of lift slope with tail geometry.

Figure 5.4.2 shows how the effective aspect ratio of the vertical stabilizer varies with the true aspect ratio of both the horizontal and vertical stabilizers for one particular tail configuration. The tail geometry used to generate the data plotted in Fig. 5.4.2 was a conventional tail similar to that shown in Fig. 5.4.1. Both the horizontal and vertical stabilizers had unswept rectangular planforms with the same chord and the horizontal stabilizer had no dihedral. The true aspect ratio of the vertical stabilizer is defined to be the square of the distance from root to tip divided by the planform area. The effective aspect ratio for the vertical stabilizer was defined to be the aspect ratio that when used in the analytical solution to Prandtl's lifting-line equation for the isolated vertical stabilizer, gives the same lift slope as that predicted by the numerical lifting-line solution for the complete tail. Strictly speaking, Fig. 5.4.2 applies only to the particular tail geometry for which it was obtained. However, in preliminary design the exact tail geometry is not typically known, and Fig. 5.4.2 can be used as a guideline.

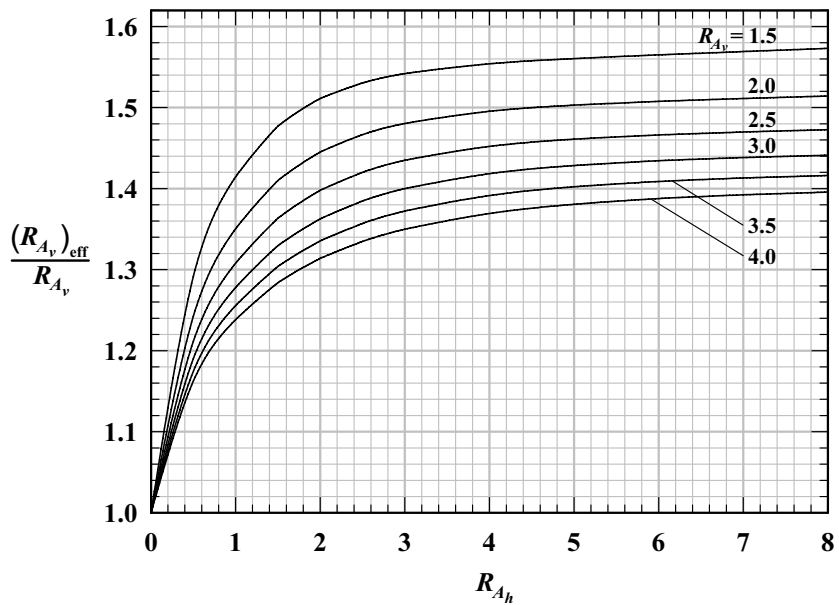


Figure 5.4.2. Effective vertical aspect ratio for a conventional tail with both the horizontal and vertical stabilizers having unswept rectangular planforms.